



IEC 61784-5-18

Edition 2.0 2018-08

INTERNATIONAL STANDARD



**Industrial communication networks – Profiles –
Part 5-18: Installation of fieldbuses – Installation profiles for CPF 18**



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INTERNATIONAL ELECTROTECHNICAL COMMISSION

INDUSTRIAL COMMUNICATION NETWORKS – PROFILES –

Part 5-18: Installation of fieldbuses – Installation profiles for CPF 18

FOREWORD

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International Standard IEC 61784-5-18 has been prepared by subcommittee 65C: Industrial networks, of IEC technical committee 65: Industrial-process measurement, control and automation.

This second edition cancels and replaces the first edition published in 2013. This edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

- a) alignment with IEC 61918:2018;
- b) addition of new connector (LC).

This document is to be used in conjunction with IEC 61918:2018.

The text of this International Standard is based on the following documents:

FDIS	Report on voting
65C/924/FDIS	65C/925/RVD

Full information on the voting for the approval of this international Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts of the IEC 61784-5 series, published under the general title *Industrial communication networks – Profiles – Installation of fieldbuses*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

IMPORTANT – The 'colour inside' logo on the cover page of this publication indicates that it contains colours which are considered to be useful for the correct understanding of its contents. Users should therefore print this document using a colour printer.

INTRODUCTION

This International Standard is one of a series produced to facilitate the use of communication networks in industrial control systems.

IEC 61918:2018 provides the common requirements for the installation of communication networks in industrial control systems. This installation profile document provides the installation profiles of the communication profiles (CP) of a specific communication profile family (CPF) by stating which requirements of IEC 61918 fully apply and, where necessary, by supplementing, modifying, or replacing the other requirements (see Figure 1).

For general background on fieldbuses, their profiles, and relationship between the installation profiles specified in this document, see IEC 61158-1.

Each CP installation profile is specified in a separate annex of this document. Each annex is structured exactly as the reference standard IEC 61918 for the benefit of the persons representing the roles in the fieldbus installation process as defined in IEC 61918 (planner, installer, verification personnel, validation personnel, maintenance personnel, administration personnel). By reading the installation profile in conjunction with IEC 61918, these persons immediately know which requirements are common for the installation of all CPs and which are modified or replaced. The conventions used to draft this document are defined in Clause 5.

The provision of the installation profiles in one document for each CPF (for example IEC 61784-5-18 for CPF 18) allows readers to work with documents of a convenient size.

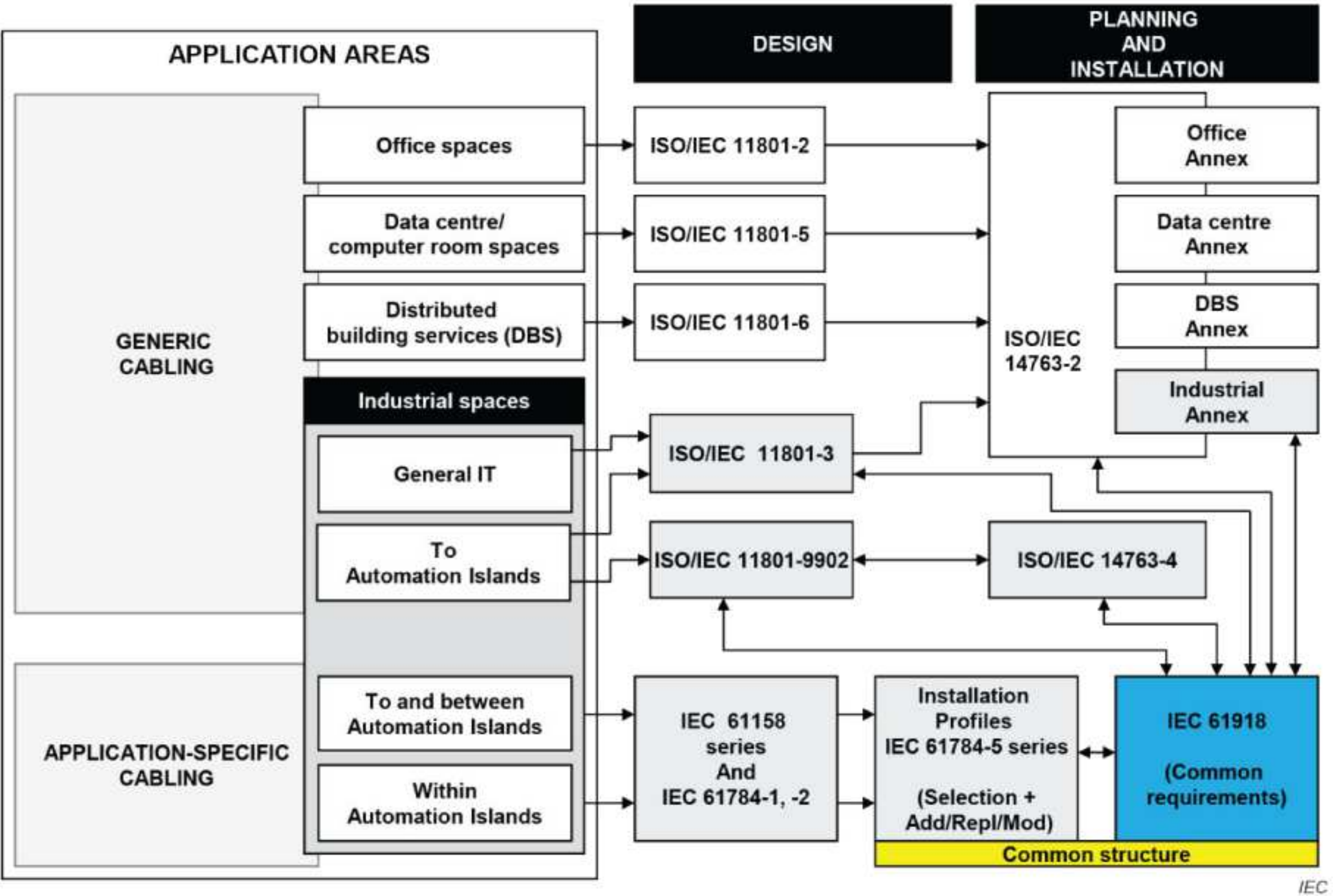


Figure 1 – Standards relationship

INDUSTRIAL COMMUNICATION NETWORKS – PROFILES –

Part 5-18: Installation of fieldbuses – Installation profiles for CPF 18

1 Scope

This part of IEC 61784-5 specifies the installation profiles for CPF 18 (SafetyNET p¹).

The installation profiles are specified in Annex A. This annex is read in conjunction with IEC 61918:2018.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 61918:2018, *Industrial communication networks – Installation of communication networks in industrial premises*

The normative references of IEC 61918:2018, Clause 2, apply.

3 Terms, definitions and abbreviated terms

For the purpose of this document, the terms, definitions and abbreviated terms given in IEC 61918:2018 Clause 3, apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

4 CPF 18: Overview of installation profiles

CPF 18 consists of two Communication Profiles as specified in IEC 61784-2.

The installation requirements for CP 18/1 (SafetyNET p RTFL) and CP 18/2 (SafetyNET p RTFN) are specified in Annex A.

¹ SafetyNET p is trade name of Pilz GmbH & Co. KG. This information is given for the convenience of users of this document and does not constitute an endorsement by IEC of the trade name holder or any of its products. Compliance to this document does not require use of the trade name SafetyNET p. Use of the trade name SafetyNET p requires permission of the trade name holder.

5 Installation profile conventions

The numbering of the clauses and subclauses in the annex of this document corresponds to the numbering of IEC 61918 main clauses and subclauses.

The annex clauses and subclauses of this document supplement, modify, or replace the respective clauses and subclauses in IEC 61918.

Where there is no corresponding subclause of IEC 61918 in the normative annexes in this document, the subclause of IEC 61918 applies without modification.

The annex heading letter represents the installation profile assigned in Clause 4. The annex heading number shall represent the corresponding numbering of IEC 61918.

EXAMPLE “Subclause A.4.4” in IEC 61784-5-18 means that CP 18/1 and CP 18/2 specify the Subclause 4.4 of IEC 61918.

All main clauses of IEC 61918 are cited and apply in full unless otherwise stated in each normative installation profile annex.

If all subclauses of a (sub)clause are omitted, then the corresponding IEC 61918 (sub)clause applies.

If in a (sub)clause it is written “Not applicable.”, then the corresponding IEC 61918 (sub)clause does not apply.

If in a (sub)clause it is written “*Addition:*”, then the corresponding IEC 61918 (sub)clause applies with the additions written in the profile.

If in a (sub)clause it is written “*Replacement:*”, then the text provided in the profile replaces the text of the corresponding IEC 61918 (sub)clause.

NOTE A replacement can also comprise additions.

If in a (sub)clause it is written “*Modification:*”, then the corresponding IEC 61918 (sub)clause applies with the modifications written in the profile.

If all (sub)clauses of a (sub)clause are omitted but in this (sub)clause it is written “(Sub)clause x has *addition:*” (or “*replacement:*”) or “(Sub)clause x is not applicable.”, then (sub)clause x becomes valid as declared and all the other corresponding IEC 61918 (sub)clauses apply.

6 Conformance to installation profiles

Each installation profile within this document includes part of IEC 61918:2018. It may also include defined additional specifications.

A statement of compliance to an installation profile of this document shall be stated² as either

Compliance to IEC 61784-5-18:— for CP 18/m <name> or

Compliance to IEC 61784-5-18 (Ed.2.0) for CP 18/m <name>

² In accordance with ISO/IEC Directives.

where the name within the angle brackets < > is optional and the angle brackets are not to be included. The m within CP 18/m shall be replaced by the profile number 1 to 2.

NOTE The name may be the name of the profile, for example SafetyNET p.

If the name is a trade name, then the permission of the trade name holder shall be required.

Product standards shall not include any conformity assessment aspects (including quality management provisions), neither normative nor informative, other than provisions for product testing (evaluation and examination).

Annex A (normative)

CP 18/1 and CP 18/2 (SafetyNET p) specific installation profile

A.1 Installation profile scope

Addition:

This annex specifies the installation profile for Communication Profile CP 18/1 and CP 18/2 (SafetyNET p). The CP 18/1 and CP 18/2 are specified in IEC 61784-2.

A.2 Normative references

A.3 Installation profile terms, definitions, and abbreviated terms

A.3.1 Terms and definitions

A.3.2 Abbreviated terms

A.3.3 Conventions for installation profiles

Not applicable.

A.4 Installation planning

A.4.1 General

A.4.1.1 Objective

A.4.1.2 Cabling in industrial premises

A.4.1.3 The planning process

A.4.1.4 Specific requirements for CPs

Not applicable.

A.4.1.5 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.2 Planning requirements

A.4.2.1 Safety

A.4.2.1.1 General

A.4.2.1.2 Electric safety

A.4.2.1.3 Functional safety

A.4.2.1.4 Intrinsic safety

Not applicable.

A.4.2.1.5 Safety of optical fibre communication systems**A.4.2.2 Security****A.4.2.3 Environmental considerations and EMC****A.4.2.4 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3****A.4.3 Network capabilities****A.4.3.1 Network topology****A.4.3.1.1 Common description****A.4.3.1.2 Basic physical topologies for passive networks**

Not applicable.

A.4.3.1.3 Basic physical topologies for active networks

Addition:

Ring topology shall be used when cable redundancy is required.

A.4.3.1.4 Combination of basic topologies**A.4.3.1.5 Specific requirements for CPs**

Not applicable.

A.4.3.1.6 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3**A.4.3.2 Network characteristics****A.4.3.2.1 General****A.4.3.2.2 Network characteristics for balanced cabling not based on Ethernet**

Not applicable.

A.4.3.2.3 Network characteristics for balanced cabling based on Ethernet

Replacement:

Table A.1 provides values based on the template given in IEC 61918:2018, Table 2.

Table A.1 – Network characteristics for balanced cabling based on Ethernet

Characteristic	CP 18/1 and CP 18/2
Supported data rates (Mbit/s)	100
Supported channel length (m) ^b	100
Number of connections in the channel (max.) ^{a,b}	6
Patch cord length (m) ^a	100
Channel class per ISO/IEC 11801-3 (min.) ^b	D
Cable category per ISO/IEC 11801-3 (min.) ^c	5
Connecting HW category per ISO/IEC 11801-3 (min.)	5
Cable types	–
^a See A.4.4.3.2.	
^b For the purpose of this table the channel definitions of ISO/IEC 11801-3 are applicable.	
^c Additional information is available in IEC 61156 series.	

A.4.3.2.4 Network characteristics for optical fibre cabling

Replacement:

Table A.2 provides values based on the template given in IEC 61918:2018, Table 3.

Table A.2 – Network characteristics for optical fibre cabling

CP 18/1 and CP 18/2		
Optical fibre type	Description	
Single mode silica	Bandwidth (MHz) or equivalent at λ (nm)	500 at 1 310
	Minimum length (m)	0
	Maximum length ^a (m)	14 000
	Maximum channel insertion loss/optical power budget (dB)	8,5
	Connecting hardware	See A.4.4.2.5
Multimode silica	Modal bandwidth (MHz x km) at λ (nm)	600 at 1 300
	Minimum length (m)	0
	Maximum length ^a (m)	2 000
	Maximum channel insertion loss/optical power budget (dB)	4,5
	Connecting hardware	See A.4.4.2.5
POF	Modal bandwidth (MHz x 100 m) at λ (nm)	35 at 650
	Minimum length (m)	0
	Maximum length ^a (m)	50
	Maximum channel insertion loss/optical power budget (dB)	14
	Connecting hardware	See A.4.4.2.5
Hard clad silica	Modal bandwidth (MHz x km) at λ (nm)	70 at 650
	Minimum length (m)	0
	Maximum length ^a (m)	100
	Maximum channel insertion loss/optical power budget (dB)	3
	Connecting hardware	See A.4.4.2.5
^a This value is reduced by connections, splices and bends in accordance with Formula (1) in 4.4.3.4.1 of IEC 61918:2018.		

- A.4.3.2.5 **Specific network characteristics**
- A.4.3.2.6 **Specific requirements for generic cabling in accordance with ISO/IEC 11801-3**
- A.4.4 **Selection and use of cabling components**
 - A.4.4.1 **Cable selection**
 - A.4.4.1.1 **Common description**
 - A.4.4.1.2 **Copper cables**
 - A.4.4.1.2.1 **Balanced cables for Ethernet-based CPs**

Replacement:

Table A.3 provides values based on the template given in IEC 61918:2018, Table 4.

Table A.3 – Information relevant to copper cable: fixed cables

Characteristic	CP 18/1 and CP 18/2
Nominal impedance of cable (tolerance) (Ω)	100 $\pm$ 15 (IEC 61156-5)
Balanced or unbalanced	Balanced
DCR of conductors (Ω /km)	$\leq$ 115 $\leq$ 62 (recommended)
DCR of shield	–
Number of conductors	4
Shielding	S/FTP, S/FTQ, S/STP
Colour code for conductor	WH, BU, YE, OG
Jacket colour requirements	Yellow (RAL 1003) (recommended)
Jacket material	No requirement Application dependent
Resistance to harsh environment (e.g. UV, oil resist, LS0H)	No requirement Application dependent
Agency ratings	No requirement

Replacement:

Table A.4 provides values based on the template given in IEC 61918:2018, Table 5.

Table A.4 – Information relevant to copper cable: cords

Characteristic	CP 18/1 and CP 18/2
Nominal impedance of cable (tolerance) (Ω)	100 $\pm$ 15 (IEC 61156-5)
Balanced or unbalanced	Balanced
DCR of conductors (Ω /km)	$\leq$ 115 $\leq$ 62 (recommended)
DCR of shield	–
Number of conductors	4
Length (m)	100
Shielding	S/FTP, S/FTQ, S/STP
Colour code for conductor	WH, BU, YE, OG
Jacket colour requirements	Yellow (RAL 1003) (recommended)
Jacket material	No requirement Application dependent
Resistance to harsh environment (e.g. UV, oil resist, LS0H)	No requirement Application dependent
Agency ratings	No requirement

A.4.4.1.2.2 Copper cables for non-Ethernet-based CPs

Not applicable.

A.4.4.1.3 Cables for wireless installation

A.4.4.1.4 Optical fibre cables

Replacement:

Table A.5 provides values based on the template given in IEC 61918:2018, Table 6.

Table A.5 – Information relevant to optical fibre cables

Characteristic	9..10/125 μ m single mode silica	50/125 μ m multimode silica	62,5/125 μ m multimode silica	980/1 000 μ m step index POF	200/230 μ m step index hard clad silica
Standard	IEC 60793-2-50 Type B1	IEC 60793-2-10 Type A1a	IEC 60793-2-10 Type A1b	IEC 60793-2-40 Type A4a	IEC 60793-2-30 Type A3c
Attenuation per km (650 nm)	–	–	–	$\leq$ 160 dB	$\leq$ 10 dB
Attenuation per km (1 310 nm)	$\leq$ 0,5 dB	$\leq$ 1,5 dB	$\leq$ 1,5 dB	–	–
Number of optical fibres	2	2	2	2	2
Jacket colour requirements	Yellow (RAL 1003) (recommended)	Yellow (RAL 1003) (recommended)	Yellow (RAL 1003) (recommended)	Yellow (RAL 1003) (recommended)	Yellow (RAL 1003) (recommended)
Jacket material	No requirement Application dependent	No requirement Application dependent	No requirement Application dependent	No requirement Application dependent	No requirement Application dependent
Resistance to harsh environment (e.g. UV, oil resist, LS0H)	No requirement Application dependent	No requirement Application dependent	No requirement Application dependent	No requirement Application dependent	No requirement Application dependent

A.4.4.1.5 Special purpose balanced and optical fibre cables

A.4.4.1.6 Specific requirements for CPs

Not applicable.

A.4.4.1.7 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.4.2 Connecting hardware selection

A.4.4.2.1 Common description

A.4.4.2.2 Connecting hardware for balanced cabling CPs based on Ethernet

Replacement:

Table A.6 provides values based on the template given in IEC 61918:2018, Table 7.

Table A.6 – Connectors for balanced cabling CPs based on Ethernet

	IEC 60603-7 series ^a		IEC 61076-3-106 ^b		IEC 61076-3-117 ^b	IEC 61076-2-101	IEC 61076-2-109
	shielded	unshielded	Var. 1	Var. 6	Var. 14	M12-4 with D-coding	M12-8 with X-coding
CP 18/1 and 18/2	IEC 60603-7-3	No	No	No	Yes	Yes	Yes
^a For IEC 60603-7 series, the connector selection is based on the desired channel performance.							
^b Housings to protect connectors.							

A.4.4.2.3 Connecting hardware for copper cabling CPs not based on Ethernet

Not applicable.

A.4.4.2.4 Connecting hardware for wireless installation

A.4.4.2.5 Connecting hardware for optical fibre cabling

Replacement:

Table A.7 provides values based on the template given in IEC 61918:2018, Table 9.

Table A.7 – Optical fibre connecting hardware

	IEC 61754-2	IEC 61754-4	IEC 61754-24	IEC 61754-20	IEC 61754-22	Others
	BFOC/2,5	SC	SC-RJ	LC	F-SMA	
CP 18/1 and CP 18/2	No	Yes	Yes	Yes	No	No
NOTE IEC 61754 series defines the optical fibre connector mechanical interfaces; performance specifications for optical fibre connectors terminated to specific fibre types are standardised in IEC 61753-1 series.						

Replacement:

Table A.8 provides values based on the template given in IEC 61918:2018, Table 10.

Table A.8 – Relationship between FOC and fibre types (CP 18/1 and CP 18/2)

	Fibre type					
	9..10/125 µm single mode silica	50/125 µm multimode silica	62,5/125 µm multimode silica	980/1 000 µm step index POF	200/230 µm step index hard clad silica	Others
SC	Yes	Yes	Yes	Yes	Yes	No
SC-RJ	Yes	Yes	Yes	Yes	Yes	No
LC	Yes	Yes	Yes	No	No	No

A.4.4.2.6 Specific requirements for CPs

Not applicable.

**A.4.4.2.7 Specific requirements for generic cabling in accordance with
ISO/IEC 11801-3**

A.4.4.3 Connections within a channel/permanent link

A.4.4.3.1 Common description

A.4.4.3.2 Balanced cabling connections and splices for CPs based on Ethernet

A.4.4.3.3 Copper cabling connections and splices for CPs not based on Ethernet

Not applicable.

A.4.4.3.4 Optical fibre cabling connections and splices for CPs based on Ethernet

**A.4.4.3.5 Optical fibre cabling connections and splices for CPs not based on
Ethernet**

Not applicable.

**A.4.4.3.6 Specific requirements for generic cabling in accordance with
ISO/IEC 11801-3**

A.4.4.4 Terminators

Not applicable.

A.4.4.5 Device location and connection

A.4.4.5.1 Common description

A.4.4.5.2 Specific requirements for CPs

Not applicable.

A.4.4.5.3 Specific requirements for wireless installation**A.4.4.5.4 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3****A.4.4.6 Coding and labelling****A.4.4.6.1 Common description****A.4.4.6.2 Additional requirements for CPs****A.4.4.6.3 Specific requirements for CPs**

Not applicable.

A.4.4.6.4 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3**A.4.4.7 Earthing and bonding of equipment and devices and shielded cabling****A.4.4.7.1 Common description****A.4.4.7.2 Bonding and earthing of enclosures and pathways****A.4.4.7.3 Earthing methods****A.4.4.7.4 Shield earthing****A.4.4.7.4.1 Non-earthing or parallel RC**

Not applicable.

A.4.4.7.4.2 Direct

Addition:

Shielding of bus cables shall always be connected to earth at both ends of the cables. Single point shield termination shall be avoided.

Where equipotential bonding is not guaranteed or cannot be achieved optical fibre cabling should be used.

A.4.4.7.4.3 Derivatives of direct and parallel RC**A.4.4.7.5 Specific requirements for CPs**

Not applicable.

A.4.4.7.6 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3**A.4.4.8 Storage and transportation of cables****A.4.4.8.1 Common description****A.4.4.8.2 Specific requirements for CPs**

Not applicable.

A.4.4.8.3 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.4.9 Routing of cables

A.4.4.10 Separation of circuits

A.4.4.11 Mechanical protection of cabling components

A.4.4.11.1 Common description

A.4.4.11.2 Specific requirements for CPs

Not applicable.

A.4.4.11.3 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.4.12 Installation in special areas

A.4.4.12.1 Common description

A.4.4.12.2 Specific requirements for CPs

Not applicable.

A.4.4.12.3 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.5 Cabling planning documentation

A.4.6 Verification of cabling planning specification

A.5 Installation implementation

A.5.1 General requirements

A.5.2 Cable installation

A.5.2.1 General requirements for all cabling types

A.5.2.1.1 Storage and installation

A.5.2.1.2 Protecting communication cables against potential mechanical damage

Replacement:

Table A.9 provides values based on the template given in IEC 61918:2018, Table 18.

Table A.9 – Parameters for balanced cables

Characteristic		Value
Mechanical force	Minimum bending radius, single bending (mm)	20 to 65 ^a
	Bending radius, multiple bending (mm)	50 to 100 ^a
	Pull forces (N)	≤ 150 ^a
	Permanent tensile forces (N)	≤ 50 ^a
	Maximum lateral forces (N/cm)	–
	Temperature range during installation (°C)	-20 to +60 ^a
^a Depending on cable type.		

Replacement:

Table A.10 provides values based on the template given in IEC 61918:2018, Table 19.

Table A.10 – Parameters for silica optical fibre cables

Characteristic		Value
Mechanical force	Minimum bending radius, single bending (mm)	50 to 200 ^a
	Bending radius, multiple bending (mm)	30 to 200 ^a
	Pull forces (N)	500 to 800 ^a
	Permanent tensile forces (N)	500 to 800 ^a
	Maximum lateral forces (N/cm)	300 to 500 ^a
	Temperature range during installation (°C)	-5 to +50 ^a
^a Depending on cable type.		

Replacement:

Table A.11 provides values based on the template given in IEC 61918:2018, Table 20.

Table A.11 – Parameters for POF optical fibre cables

Characteristic		Value
Mechanical force	Minimum bending radius, single bending (mm)	30 to 100 ^a
	Bending radius, multiple bending (mm)	50 to 150 ^a
	Pull forces (N)	50 to 100 ^a
	Permanent tensile forces (N)	Not allowed
	Maximum lateral forces (N/cm)	35 to 100 ^a
	Temperature range during installation (°C)	0 to 50 ^a
^a Depending on cable type.		

Replacement:

Table A.12 provides values based on the template given in IEC 61918:2018, Table 21.

Table A.12 – Parameters for hard clad silica optical fibre cables

Characteristic		Value
Mechanical force	Minimum bending radius, single bending (mm)	75 to 200 ^a
	Bending radius, multiple bending (mm)	75 to 200 ^a
	Pull forces (N)	100 to 800 ^a
	Permanent tensile forces (N)	≤ 100 ^a
	Maximum lateral forces (N/cm)	≤ 75 to 300 ^a
	Temperature range during installation (°C)	-5 to +50 ^a
^a Depending on cable type.		

- A.5.2.1.3 Avoid forming loops**
- A.5.2.1.4 Torsion (twisting)**
- A.5.2.1.5 Tensile strength (on installed cables)**
- A.5.2.1.6 Bending radius**
- A.5.2.1.7 Pull force**
- A.5.2.1.8 Fitting strain relief**
- A.5.2.1.9 Installing cables in cabinet and enclosures**
- A.5.2.1.10 Installation on moving parts**
- A.5.2.1.11 Cable crush**
- A.5.2.1.12 Installation of continuous flexing cables**
- A.5.2.1.13 Additional instructions for the installation of optical fibre cables**
- A.5.2.2 Installation and routing**
- A.5.2.3 Specific requirements for CPs**

Not applicable.

- A.5.2.4 Specific requirements for wireless installation**
- A.5.2.5 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3**
- A.5.3 Connector installation**
 - A.5.3.1 Common description**
 - A.5.3.2 Shielded connectors**
 - A.5.3.3 Unshielded connectors**
 - A.5.3.4 Specific requirements for CPs**

Not applicable.

- A.5.3.5 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3**

A.5.4 Terminator installation

Not applicable.

A.5.5 Device installation

- A.5.5.1 Common description**
- A.5.5.2 Specific requirements for CPs**

Not applicable.

A.5.6 Coding and labelling**A.5.6.1 Common description****A.5.6.2 Specific requirements for CPs**

Not applicable.

A.5.7 Earthing and bonding of equipment and devices and shield cabling**A.5.7.1 Common description****A.5.7.2 Bonding and earthing of enclosures and pathways****A.5.7.3 Earthing methods****A.5.7.4 Shield earthing methods****A.5.7.5 Specific requirements for CPs**

Not applicable.

A.5.7.6 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3**A.5.8 As-implemented cabling documentation****A.6 Installation verification and installation acceptance test****A.6.1 General****A.6.2 Installation verification****A.6.2.1 General****A.6.2.2 Verification according to cabling planning documentation****A.6.2.3 Verification of earthing and bonding****A.6.2.4 Verification of shield earthing****A.6.2.5 Verification of cabling system****A.6.2.6 Cable selection verification****A.6.2.6.1 Common description****A.6.2.6.2 Specific requirements for CPs**

Not applicable.

A.6.2.6.3 Specific requirements for wireless installation**A.6.2.7 Connector verification****A.6.2.7.1 Common description****A.6.2.7.2 Specific requirements for CPs**

Not applicable.

A.6.2.7.3 Specific requirements for wireless installation

A.6.2.8 Connection verification

A.6.2.9 Terminator verification

Not applicable.

A.6.2.10 Coding and labelling verification

A.6.2.10.1 Common description

A.6.2.10.2 Specific coding and labelling verification requirements

Not applicable.

A.6.2.11 Verification report

A.6.3 Installation acceptance test

A.6.3.1 General

A.6.3.2 Acceptance test of Ethernet-based cabling

A.6.3.2.1 Validation of balanced cabling for CPs based on Ethernet

A.6.3.2.1.1 Common description

A.6.3.2.1.2 Transmission performance test parameters

A.6.3.2.1.3 Specific requirements for CPs based on Ethernet

Not applicable.

A.6.3.2.2 Validation of optical fibre cabling for CPs based on Ethernet

A.6.3.2.2.1 Common description

A.6.3.2.2.2 Specific requirements for optical fibre cabling CPs

Not applicable.

A.6.3.2.3 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.6.3.3 Acceptance test of non-Ethernet-based cabling

Not applicable.

A.6.3.4 Specific requirements for wireless installation

A.6.3.5 Acceptance test report

A.7 Installation administration

A.7.1 General

A.7.2 Fields covered by the administration

A.7.3 Basic principles for the administration system

A.7.4 Working procedures

A.7.5 Device location labelling

A.7.6 Component cabling labelling

A.7.7 Documentation

A.7.8 Specific requirements for administration

Not applicable.

A.8 Installation maintenance and installation troubleshooting

A.8.1 General

A.8.2 Maintenance

A.8.3 Troubleshooting

A.8.4 Specific requirements for maintenance and troubleshooting

Not applicable.

Bibliography

Addition:

- [40] IEC 61158-3-22, *Industrial communication networks – Fieldbus specifications – Part 3-1: Data-link layer service definition – Type 22 elements*
 - [41] IEC 61158-4-22, *Industrial communication networks – Fieldbus specifications – Part 4-1: Data-link layer protocol specification – Type 22 elements*
 - [42] IEC 61158-5-22, *Industrial communication networks – Fieldbus specifications – Part 5-5: Application layer service definition – Type 22 elements*
 - [43] IEC 61158-6-22, *Industrial communication networks – Fieldbus specifications – Part 6-5: Application layer protocol specification – Type 22 elements*
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